



Towards efficient wafer visual inspection: Exploring novel lightweight approaches for anomaly detection and defect segmentation

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ABSTRACT

The rapid advancement of both wafer manufacturing and AI technologies is reshaping the semiconductor industry. As chip features become smaller and more intricate, the variety and complexity of defects continue to grow, making defect detection increasingly challenging. Meanwhile, AI has made significant strides in unsupervised anomaly detection and supervised defect segmentation, yet its application to wafer inspection remains underexplored. This work bridges these fields by investigating cutting-edge lightweight AI techniques for automated inspection of current generation of silicon wafers. Our study leverages a newly curated dataset comprising 1,055 images of 300 mm wafers, annotated with 6,861 defect labels across seven distinct types, along with PASS/FAIL decisions. From a data-centric perspective, we introduce a novel unsupervised dataset-splitting approach to ensure balanced representation of defect classes and image features. Using the DINO-ViT-S/8 model for feature extraction, our method achieves 96% coverage while maintaining the target 20% test ratio for both individual defects and PASS/FAIL classification. From a model-centric perspective, we benchmark several recent methods for unsupervised anomaly detection and supervised defect segmentation. For unsupervised anomaly detection, EfficientAD obtains the best performance for both pixel-level and image-wise metrics, with F1-scores of 75.14% and 82.35%, respectively. For supervised defect segmentation, UPerNet-Swin achieves the highest performance, with a pixel-level mDice of 47.90 and a mask-level F1-score of 57.45. To facilitate deployment in high-throughput conditions, we conduct a comparative analysis of computational efficiency. Finally, we explore a dual-stage output fusion approach that integrates the best-performing unsupervised anomaly detection and supervised segmentation models to refine PASS/FAIL decisions by incorporating defect severity.

1. Introduction

Silicon wafers are the foundation of modern electronics manufacturing, enabling the production of advanced semiconductor chips essential for devices like smartphones, computers, and communication systems. As electronic integrated circuits evolve and the demand for high-performance chips continues to grow, process precision has reached sub-10 nm scales (Chen, Zhang, Cheng, Gomez, & Gely, 2023), with ongoing advancements driving further miniaturization. However, this trend also introduces increased diversity and complexity of surface defects on wafers, creating significant challenges for defect detection techniques in semiconductor manufacturing (Ma et al., 2023).

Artificial intelligence (AI) is transforming the industrial sector, revolutionizing quality assurance processes in electronics manufacturing. Visual inspection, one of AI's primary applications, plays a critical role in ensuring defect-free products. Traditionally, wafer defect inspection relied on human operators or rule-based computer vision systems, but

these approaches face limitations such as low efficiency, inconsistent accuracy, high costs, and subjective judgment. Modern methods leveraging machine vision, particularly Computer Vision (CV) and Deep Learning (DL), have increasingly replaced manual inspection, offering improved detection accuracy, scalability, and reproducibility.

In recent years, anomaly detection has gained significant traction as an approach for visual inspection in scenarios with limited labeled defect data. This method seeks to identify deviations from expected patterns, operating under an open-set assumption that enables the model to recognize defects that are unknown or previously unseen. This is particularly valuable in the semiconductor industry, where processes are optimized to minimize defects, making it easy to obtain normal samples but challenging to collect a comprehensive set of defect examples. Consequently, anomaly detection can help manufacturers detect novel or emerging anomalies, functioning as a flexible first line of inspection in rapidly evolving production environments. However, anomaly detection systems typically lack the ability to classify

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specific defect types, nor can they distinguish between actual defects and benign variations in the manufacturing process. While effective at flagging outliers, additional analysis is often required to interpret the root causes of detected anomalies, particularly in contexts with high variability in defect characteristics.

In contrast, defect segmentation approaches adopt a closed-set framework, trained on a predefined set of defect types to accurately localize and classify known flaws. This approach is valuable for semiconductor manufacturing, where specific, known defect classes must be reliably identified to ensure quality control. Defect segmentation systems produce pixel-wise segmentation maps, precisely delineating the boundaries and classes of detected defects. The main advantage of this approach lies in its ability to classify defects with a high degree of specificity, reducing false positives and enhancing the consistency of inspection outcomes. However, its performance is constrained by the quality and diversity of labeled defect samples available for training, which limits its effectiveness in identifying novel defect types that were not represented in the training data.

Both wafer manufacturing and AI technologies are advancing rapidly. In wafer fabrication, chip features are becoming smaller and more intricate, while the variety and complexity of defect types continue to grow (Ma et al., 2023). Concurrently, advancements in AI – both in supervised and unsupervised methods, such as novel anomaly detection and defect segmentation models – are achieving remarkable progress in performance and lightweight designs, thus promoting its adaptability and suitability to complex and dynamic monitoring environments. This work aims to bridge these two rapidly advancing fields by investigating the potential of cutting-edge AI techniques for automated inspection of the latest generation of silicon wafers, characterized by increasing complexity and miniaturization. Specifically, from a data-centric perspective, we propose a novel unsupervised dataset-splitting approach to ensure a balanced representation across splits, meaning to produce representative train/test splits with equal distributions of both defect classes and image features. This work was supported by a newly acquired dataset comprising 1055 images of cropped regions from various 300 mm wafers, each containing 8 mm dies of the same product. The images have a resolution of 992×992 pixels and a spatial resolution of $3.4 \mu\text{m}/\text{pixel}$. The dataset includes 6861 pixel-wise mask annotations spanning seven distinct defect types and image-wise PASS/FAIL annotations for each image, all manually labeled by six human annotators with experience in wafer defect inspection. From a model-centric perspective, we benchmark several state-of-the-art methods for unsupervised anomaly detection and supervised defect segmentation, tailored to the challenges of wafer defect detection. Additionally, to further support the decision-making process, we explore a dual-stage output fusion approach that integrates the outputs of the best-performing unsupervised and supervised methods. By integrating both strategies, this fusion refines and complements the PASS/FAIL decision from the anomaly detection module by considering the severity of various defect types detected by the semantic segmentation module within the same image.

2. Related work

Automated approaches utilizing AI for wafer inspection in electronics manufacturing have gained significant traction, particularly in computer vision. These methods encompass conventional image processing techniques as well as supervised and unsupervised Machine Learning (ML) approaches for detecting defects on wafer surfaces.

Several prior studies have explored wafer inspection techniques based purely on traditional image processing. These methods include wavelet transform (Yang, 2009; Yeh, Wu, Ji, & Huang, 2010), spatial filtering (Ohshige et al., 1991; Wang, 2008), and template matching (Barone, 2020). While these techniques offer interpretable solutions, they often struggle with generalizing across varying defect types and complex surface textures.

Supervised ML-based wafer inspection methods can be categorized into conventional ML and DL approaches. Conventional ML techniques for wafer defect detection have employed classifiers such as k-Nearest Neighbor (kNN) (Yuan, Kuo, & Bae, 2011), Decision Trees (Piao, Jin, Lee, & Byun, 2018), and Support Vector Machines (SVM) (Baly & Hajji, 2012; Xie, Huang, Gu, & Cao, 2014). While these methods have shown effectiveness in detecting known defect types, they are often limited by their reliance on hand-crafted features. Regarding DL, convolutional neural networks (CNNs) have been extensively used for wafer inspection. Significant research has been conducted on CNN-based image classification to distinguish between defective and non-defective wafers (Kyeong & Kim, 2018; Nakazawa & Kulkarni, 2018; Wang, Xu, Yang, Zhang, & Li, 2020). Additionally, DL-based object detection approaches, including Faster R-CNN, RetinaNet, YOLO, and Mask R-CNN, have been explored to identify specific defect regions (Chen, Kang, & Perng, 2020; Shinde, Pai, & Adiga, 2022; Wang, Jia, Jiang, & Jiang, 2022). More recently, semantic segmentation techniques have gained attention for segmenting individual defects, utilizing models such as U-Net, FCN, SegNet, WaferSegClassNet, and YOLOSeg (Han et al., 2020; Li, Chan, Huang, Hsu, & Chen, 2025; Nag, Makwana, Mittal, & Mohan, 2022).

However, most of these studies focus on wafer-level defect pattern analysis rather than detecting and segmenting specific defect types (e.g., scratches, contamination, particles) on individual dies at high micron/pixel resolution. This level of granularity is crucial for wafer quality assessments, as it allows for precise PASS/FAIL evaluations and facilitates root cause analysis based on defect size and shape. Additionally, the impact of newer state-of-the-art semantic segmentation architectures on defect detection in high-resolution die images remains largely unexplored. Recent models such as SegFormer (SegFormer-B0) (Xie et al., 2021), UPerNet with Swin Transformer backbone (UPerNet-Swin) (Liu et al., 2021), or Proportional-Integral-Derivative Network (PIDNet) (Xu, Xiong, & Bhattacharyya, 2023) offer promising advancements in multi-scale defect representation and real-time performance, yet their application in wafer defect segmentation is under-researched (Mei, Luo, Qiao, & Chen, 2024).

Unsupervised ML techniques have also been investigated for wafer inspection, primarily through clustering methods such as Kohonen's Self-Organizing Feature Maps (SOM), DBSCAN, and Gaussian Mixture Models (Jin, Na, Piao, Pok, & Ryu, 2019; Palma, Nicolao, Miraglia, Pasquinetti, & Piccinini, 2005). Similar to supervised methods, most of these approaches focus on wafer maps rather than high-resolution images of individual dies. More recently, DL-based anomaly detection methods, which detect defects as deviations from learned normal patterns, have gained attention. Techniques such as Coupled-hypersphere-based Feature Adaptation (CFA) (Lee, Lee, & Song, 2022), FastFlow (Yu et al., 2021), or EfficientAD (Batzner, Heckler, & König, 2024) have demonstrated strong performance on public benchmark datasets like MVTec (Bergmann, Fauser, Sattlegger, & Steger, 2019) and ViSA (Zou, Jeong, Pemula, Zhang, & Dabeer, 2022). This new generation of anomaly detection methods is particularly appealing for its efficiency in memory usage and computational speed, yet its application to wafer inspection remains relatively unexplored (Hasan, Yu, & Khan Mirani, 2024; Vieira e Silva et al., 2024).

Thus, in this work, we aim to advance the field of automated wafer visual inspection by exploring and benchmarking state-of-the-art methods for unsupervised anomaly detection and supervised defect segmentation. Specifically, we focus on detecting anomalies and defects in high-resolution individual die images, a modality that has been relatively underexplored in the literature. Additionally, we investigate hybrid strategies that combine unsupervised and supervised approaches, leveraging the complementary strengths of both methodologies. Furthermore, we enhance data preparation strategies, particularly by optimizing dataset splitting methods for representative multilabel stratified sampling. By addressing these gaps, this study seeks to contribute valuable insights into improving the robustness and efficiency of AI-driven wafer inspection techniques.

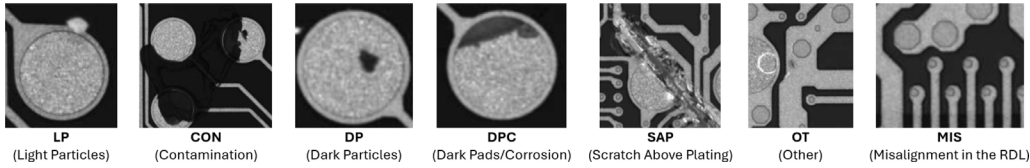


Fig. 1. Illustrative examples of annotated defect types for wafer inspection.

3. Dataset and data preparation

This section describes the image dataset used and the corresponding data preparation procedures. The dataset was created to address the lack of publicly available image datasets for the latest generation of silicon wafers, including annotations for PASS/FAIL decisions for each die and segmentation of various defect types.

Wafer images in this work were acquired under controlled industrial conditions with consistent lighting, focus, and positioning, ensuring uniform and reliable visual data for segmentation and anomaly detection. In contrast, non-controlled environments often introduce variability such as poor illumination or motion blur, requiring image quality metrics (Zhou et al., 2023, 2025) or postprocessing (e.g., denoising, gamma correction) (Yang, Chen, Zhou, Li, & Peng, 2020; Zhou, Wu et al., 2024) before the data can be reliably used in machine learning pipelines. Thanks to the precision of our acquisition setup, such corrections were unnecessary in our case.

3.1. Dataset

The acquired dataset comprises 1055 images of cropped regions from various 300 mm wafers, each containing 8 mm dies of the same product. The images have a resolution of 992×992 pixels and a spatial resolution of $3.4 \mu\text{m}/\text{pixel}$. The dataset includes 6861 pixel-wise mask annotations of distinct defect types and image-wise PASS/FAIL annotations for each image, all manually labeled by six human annotators with experience in wafer defect inspection. In particular, it includes seven distinct defect types (see Fig. 1): *LP* (Light Particles), *CON* (Contamination), *DP* (Dark Particles), *DPC* (Dark Pads/Corrosion), *SAP* (Scratch Above Plating), *OT* (Other), and *MIS* (Misalignment in the RDL). It is worth noting that the *OT* class exhibits high intra-class variability, encompassing defects with distinct visual characteristics, such as bubbles and missing copper plating, which stems from their limited representation in the dataset.

For image-wise annotations, the dataset includes 446 PASS images and 609 FAIL images. Table 1 provides the object counts for each defect type in the dataset, along with the number of annotated defects per PASS/FAIL decision. To provide further insights regarding the volume of annotations of a specific defect per image, an analysis of the occurrence of these defects at the image level is also provided, being counted the number of images in which a target defect was annotated at least one time.

DPC is the most common defect type in the dataset due to its tendency to cover large areas of the wafer and affect multiple pads simultaneously, leading to a high number of annotations. This is further supported by the defect counts at the image level, where a relatively small number of images contain this type of defect. Similarly, though more pronounced, the *MIS* defect type also has a high number of object annotations despite being present in only 11 images, as these defects often encompass many defects in the same image. Furthermore, while the PASS/FAIL annotation distribution can be considered balanced, only 15 images in the dataset lack defect annotations. This creates a unique challenge for anomaly detection while providing a valuable opportunity to explore strategies for modeling normality in wafer inspection under such imbalanced conditions.

Table 1

Dataset annotation counts per decision. Object-level annotations (left) where each object is counted as a single annotation and Image-level annotations (right) which accounts for defects that appear in an image at least once.

Defect type	Object-level counts			Image-level counts		
	PASS	FAIL	Total	PASS	FAIL	Total
LP	763	287	1050	307	265	572
CON	33	310	343	21	245	266
DP	294	118	412	203	124	327
DPC	1284	3283	4567	43	104	147
SAP	6	168	174	2	150	152
OT	5	29	34	5	24	29
MIS	65	216	281	4	7	11
NONE	–	–	–	15	0	15

3.2. Dataset splitting

Ensuring proper representation of features in train/test splits is crucial in ML to prevent bias, improve generalization, and ensure the model learns patterns that accurately reflect real-world data. The commonly used random splitting approach often results in an imbalanced distribution of annotated classes between the training and test sets, especially in low-data scenarios. Furthermore, non-annotated image characteristics – such as variations in contrast, brightness, or structural elements – can also significantly impact model performance. In the context of unsupervised anomaly detection, where models learn the normal data distribution to identify anomalies as deviations from this, it is essential to ensure that both annotated information and image characteristics are evenly represented across the train and test sets. During this work, we developed a tool that ensures this representation across sets. Additionally, an accompanying interface (see Appendix C) was also developed to extract more insights from the dataset, as well as the resulting splits.

Given the multi-decision and multi-label nature of the dataset, we first used a previously proposed iterative stratification approach (Sechidis, Tsoumakas, & Vlahavas, 2011) to effectively split the dataset into train and test sets. This method ensures a balanced representation of PASS/FAIL decision and defect types across the splits. The chosen *MultilabelStratifiedShuffleSplit* (Sechidis et al., 2011) works by grouping the dataset into labeled sets of data samples with the same labels and tries to balance the number of instances in each label set to the desired proportion in the train and test splits. Following this, the secondary objective is to ensure that the visual characteristics present in the training set are also well-represented in the test set, thereby minimizing visual biases in the final model. Vision Transformers (ViTs) have demonstrated their ability to extract meaningful features from image datasets without requiring specific adaptations to the target dataset (Caron et al., 2021). Based on this, we employed a DINO-ViT-S/8 model to extract unsupervised features, yielding rich and informative representations of the dataset images. This compression facilitates the computations to determine image proximity.

Building on the previously proposed *Improved-Precision/Recall* (Kynkäänniemi, Karras, Laine, Lehtinen, & Aila, 2019) methodology for assessing the representativeness of synthetic data relative to real data, we applied this concept to evaluate how effectively the different data splits capture the variability of the available image characteristics of

the entire dataset, ensuring that each split reliably represents the data distribution.

We begin by estimating the manifold of the test set using a k -NN strategy in the feature space produced by the vision transformer model. For each test sample, we compute the distance to its k th nearest neighbor (in this case, the third nearest), forming a hypersphere that defines its local neighborhood or area of influence. The union of all these hyperspheres provides a tractable approximation of the test manifold. The next step is to determine if the train samples are covered by this test manifold, which is done by computing the distance between a train sample to every other test sample. If the train sample is closer to at least one of the test samples than its k th nearest neighbor, then it is inserted in the hypersphere of the test sample, and is therefore covered by the test set.

Algorithm 1 demonstrates the proposed unsupervised dataset-splitting approach. It employs a simple greedy search algorithm aimed at maximizing the coverage score for the consecutive train/test folds generated by the iterative stratification k -fold splitter. At the end of this procedure, the dataset split with the highest coverage score is selected as the most representative. The corresponding train/test instance indexes are provided, ensuring adherence to the stratification criteria of the multilabel splitter while maximizing coverage for visual features. Further details about the application of the proposed approach to split the collected wafer dataset are provided on Appendix A, with a pre-defined test set size of 20%. In addition to this, we studied the robustness of this solution in Appendix B.

Algorithm 1: Representative Splitting Algorithm

Data: Dataset D , Test size s_{test} , Split Folds n_{folds} , Target Classes T_C , Vision Model $m(\theta)$

Result: Train Set D_{train} , Test Set D_{test}

$D_m = \text{merge_attributes}(D, T_C);$ /* merge target attributes into one */

$L_m = \text{binarize_labels}(D_m);$ /* convert labels to sparse matrix */

$\text{Splitter} = \text{MultilabelStratifiedShuffleSplitter}(n_{folds}, s_{test});$

$F = \text{extract_features}(m(\theta), D);$ /* reduce dimensionality of images */

$best = -\infty;$

$split_{idx} = NULL;$

for $(train_{idx}, test_{idx})$ **in** $\text{Splitter}(L_m)$ **do**

$F_{train} = F[train_{idx}];$

$F_{test} = F[test_{idx}];$

$score = \text{ImprovedRecall}(F_{train}, F_{test});$

if $score > best$ **then**

$best = score;$

$split_{idx} = test_{idx}$

else

 pass

end

end

$D_{train} = D[\neg split_{idx}];$ /* select train samples */

$D_{test} = D[split_{idx}];$ /* select test samples */

4. Methodology

The purpose of this work is to develop a solution that can classify a single image as containing an anomaly, and additionally, to detect specific defects within the image. Despite recent advances in and proven efficiency and versatility of object detection (Cheng et al., 2024; Song et al., 2025; Zhang, Guo, Jiao, Zhang and Han, 2023), some of the instances in the dataset have defects comprising the entire image, which would result in bounding boxes encompassing the entire image. For this reason, we opted to use semantic segmentation models for the classification of specific defects.

Our approach explores and benchmarks several lightweight state-of-the-art methods for unsupervised anomaly detection and supervised defect segmentation, areas that remain largely unexplored in wafer inspection. Furthermore, we investigate hybrid strategies that combine the best-performing unsupervised anomaly detection and supervised segmentation models, refining PASS/FAIL decisions by incorporating defect severity.

4.1. Anomaly detection

Three different state-of-the-art unsupervised anomaly detection models were explored to identify wafer dies that should be discarded due to anomalies. These models learn the normal data distribution and detect anomalies as deviations; therefore, only instances of the training set with PASS image-wise annotation were used during training:

- **CFA** (Lee et al., 2022): This approach mitigates bias in anomaly localization from pre-trained CNNs, which typically approximate normal feature distributions without adapting to target datasets, potentially overestimating abnormal features' normality. Therefore, CFA (Lee et al., 2022) approach was introduced, which achieves sophisticated anomaly localization by adapting features to the target dataset. CFA comprises two components: (1) a trainable patch descriptor that learns and incorporates target-oriented features, and (2) a scalable memory bank that is not dependent on the size of the target dataset. Moreover, CFA employs transfer learning to enhance the density of normal features, enabling clear differentiation of abnormal features by applying the patch descriptor and memory bank to a pre-trained CNN.
- **FastFlow** (Yu et al., 2021): Unsupervised anomaly detection often fails to map image features to a tractable base distribution and overlooks local-global feature relationships. FastFlow addresses this by extracting representative features from input images using a ResNet or Vision Transformers. For a ResNet backbone, like the one used for this paper's experiments, features from the last layer in the first three blocks are selected and fed into three flow normalization models. During the training phase, FastFlow learns to transform the input visual feature into a tractable distribution by using 2D flow normalization. During the inference phase, the features of anomalous images are expected to deviate from the distribution of normal images, resulting in lower likelihoods. This likelihood serves as the anomaly score. To obtain the final probability map, the two-dimensional probabilities of each channel are aggregated and upsampled to the input image resolution.
- **EfficientAD** (Batzner et al., 2024): Previous architectures typically use DL models as image feature extractors, leading to higher latency in implementations. To address this, Batzner et al. introduced a compact Patch Description Network (PDN) that distills knowledge from a pre-trained WideResNet-101 model. This network has the added advantage of producing localized feature vectors, minimizing the impact of anomalies in one part of an image on the rest. The PDN is then used within a teacher-student framework, where the student network is trained to replicate the teacher's output on normal images while being penalized to prevent imitation on out-of-distribution ImageNet images. Additionally, an autoencoder network is employed to identify logical errors in images. Since autoencoders often struggle to generate clear images, the student network is trained to learn the autoencoder's output for normal images. The resulting averages of the student predictions are then combined into a final heatmap to determine the anomaly score and localize anomalies.

These models were selected due to their proved performance in public industrial anomaly detection datasets like MVTec (Bergmann et al., 2019) and ViSA (Zou et al., 2022). Despite these datasets not having specific subsets comprising wafer images, there are some images

that pertain to electrical components, such as MVTec's *transistors* and ViSA's *PCB* subsets. Another selection criterion was their efficiency regarding memory usage and latency, as demonstrated in published benchmarks (Batznner et al., 2024; Lee et al., 2022; Yu et al., 2021). For this reason, diffusion-based models and other foundational models were outright discarded from the pool (Kwak et al., 2024; Mousakhan, Brox, & Tayyub, 2023; Tebbe & Tayyub, 2024; Zhang, Wang, Wu and Jiang, 2023). Other zero/few-shot alternatives, such as WinCLIP (Jeong et al., 2023) and AnomalyCLIP (Zhou, Pang, Tian, He and Chen, 2024) were discarded due to their slightly higher overhead and difficulty in modeling the normality of highly diverse datasets such as the one being used. For each model, we applied the same hyperparameters as reported in the corresponding source papers. A batch size of 8 was applied to all models, except for EfficientAD, which only supports processing one image at a time. Additionally, replacing FastFlow's default Adam optimizer with AdamW and incorporating a cosine annealed learning rate scheduler significantly improved model performance and convergence.

In addition to these models, we compare against the Six-Sigma approach to assess how modern algorithms outperform traditional statistical methods. Six-Sigma identifies anomalies as data points that fall outside a range defined by the mean and a multiple of the standard deviation (typically $\pm 3\sigma$ or $\pm 6\sigma$), assuming a normal distribution. This simple and interpretable method is effective for detecting outliers in datasets with stable statistical properties, such as pixel values or sensor readings.

4.2. Defect segmentation

Three recent supervised semantic segmentation models available on the MMsegmentation framework (Contributors, 2020) were explored to detect seven different types of defects in wafer dies, utilizing the pixel-wise mask annotations from the collected dataset:

- **SegFormer-B0** (Xie et al., 2021) integrates the lightweight Mix Transformer-B0 (MiT) encoder with a multi-layer perceptron (MLP) decoder. The hierarchical transformer encoder captures multi-scale features with a larger effective receptive field compared to traditional CNNs, while the lightweight MLP decoder efficiently combines local and global information. This makes SegFormer particularly suitable for capturing defects of varying sizes and complexities. Its flexibility in scaling, based on computational resources, ensures its applicability across different hardware setups.
- **UPerNet-Swin** (Liu et al., 2021) combines UPerNet's powerful feature fusion for multi-scale context aggregation with the Swin Transformer's efficient hierarchical attention mechanism. UPerNet excels in integrating high and low-resolution features for detailed segmentation, while the shifted window approach of the Swin Transformer captures both local and global features with computational efficiency. Together, this architecture provides robust and scalable performance for dense prediction tasks like defect segmentation.
- **PIDNet** (Xu et al., 2023) employs a unique three-branch architecture tailored for detail, context, and boundary refinement. While traditional two-path designs often blur boundaries by merging low-level and high-level features, PIDNet's boundary attention branch explicitly captures high-frequency features to sharpen boundaries. The fusion of the detail, context, and boundary branches ensures precise segmentation, particularly for defects where boundary clarity is critical. Additionally, PIDNet's design prioritizes computational efficiency, making it ideal for real-time applications.

These models were chosen based on their diverse design principles, relevant performance in related tasks, and suitability for the unique challenges posed by the wafer inspection use case, such as multi-scale defect representation, boundary precision, and real-time performance requirements. The experiments conducted in this study utilized Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019), a framework for automatic hyperparameter optimization, to effectively calibrate the model parameters and improve their overall performance. This automated approach explored a wide range of learning rates ($1e-5$ to $1e-3$), batch sizes (4, 8, 16, and 32), and optimizers (Adam and AdamW). Each architecture was optimized over 20 trials, ensuring well-calibrated models for segmentation tasks.

4.3. Dual-stage output fusion approach

Previous sub-sections focused on assessing how novel lightweight approaches for anomaly detection and defect segmentation independently optimize automatic wafer inspection. Here, we extend this analysis by evaluating the advantages and limitations of applying a non-trainable dual-stage approach that integrates the best-performing unsupervised and supervised methods. Different trainable fusion strategies can be applied for that purpose, like feature-level fusion strategies that combine extracted feature maps from both models using concatenation or attention, and then process them through a final classifier head. While trainable approaches offer deeper integration, they also present challenges such as increased computational complexity, reduced interpretability (as feature fusion often leads to a black-box effect), and potential compatibility issues (e.g., incompatible feature spaces arising from different architectures). On the other hand, non-trainable approaches like output fusion offer several benefits for scenarios with lightweight and high-throughput requirements, such as improved computational efficiency and better interpretability, as individual model decisions remain transparent.

Given this, in this work, we chose to combine both strategies at the output level using a non-trainable strategy. Our goal is to refine and complement the PASS/FAIL decision from the anomaly detection module, in particular by incorporating the severity of the defect types detected by the semantic segmentation module. For that purpose, defects can be grouped into two main categories based on the defect severity, specifically their impact on the functionality and performance of the die:

- **Critical Defects** (Leading to FAIL/Discard): These defects directly impact the functionality, structural integrity, or reliability of the die, making them non-recoverable. They pose a significant risk to downstream processes and end-product performance, often leading to immediate rejection. Defect types like CON and SAP usually fall into this category.
- **Non-Critical Defects** (Leading to PASS/Recoverable): These defects, while visible, do not necessarily impact the die's functionality and can either be tolerated within specifications or corrected through rework processes. Defect types like LP, DP, DPC, MIS, and OT can often be included in this category.

By complementing the PASS/FAIL decision from the anomaly detector at the image-level, with the critical/non-critical severity of the defects detected on the same image, we can obtain 5 possible aggregated classes: (i) PASS decision without any defects ($PASS_{NONE}$); (ii) PASS decision with only non-critical defects ($PASS_{RECOV}$); (iii) PASS decision with at least one critical defect ($PASS_{DISCARD}$); (iv) FAIL decision with only non-critical defects ($FAIL_{RECOV}$); and (v) FAIL decision with at least one critical defect ($FAIL_{DISCARD}$). The performance of the integrated approach for the automated detection of these five aggregated classes will be analyzed, alongside a discussion of its benefits and limitations. This will highlight how the approach can support actionable decision-making, particularly in human-in-the-loop scenarios.

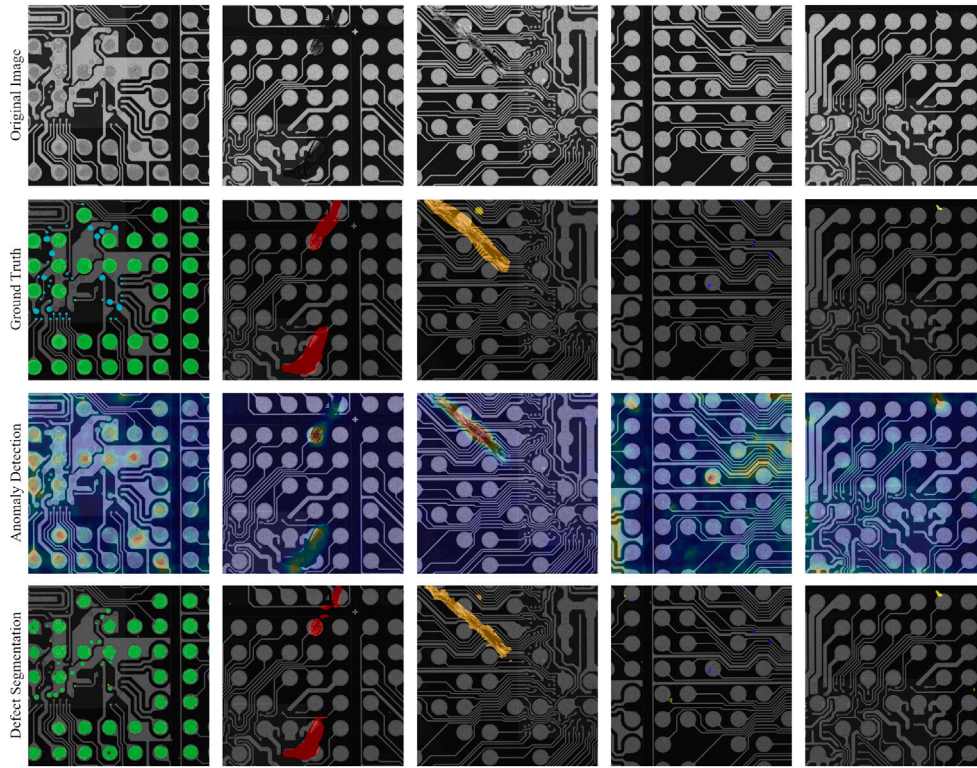


Fig. 2. Illustrative examples of the obtained results. Anomaly Detection predictions were obtained using the EfficientAD model, while Defect Segmentation predictions were produced with the UPerNet-Swin model. Defect types are color-coded as follows: green for DPC, red for CON, orange for SAP, blue for DP, and yellow for LP.

5. Results and discussion

This section presents and discusses the results separately for unsupervised anomaly detection and supervised defect segmentation (see Fig. 2), as well as for a dual-stage output fusion approach that integrates the outputs of the best-performing models from both tasks. Additionally, to evaluate the suitability of these approaches for deployment in high-throughput real-world scenarios, a comparative analysis of their computational efficiency is provided.

5.1. Anomaly detection

In this section, the results of the benchmarks with the unsupervised anomaly detection models will be explained. The outputs of these models are usually anomaly/heat maps that indicate the location and the intensity of the anomaly in the image (see Fig. 2). The confidence of the model is obtained by computing the pixel with the highest value in the entire map, which can later be used to determine the binary decision. Following this logic, we chose two types of metrics, to evaluate the model performance on the image-level decision and pixel-wise values, which we can compare with the ground truth defect segmentations.

5.1.1. Classification threshold selection

The primary objective of this component is to minimize the likelihood of an anomalous sample being propagated through the production line. To achieve this, it is crucial to reduce the model's miss rate, or False Negative Rate (FNR), as much as possible. At the same time, some consideration must be given to the False Positive Rate (FPR) to ensure the model does not become overly biased towards the anomaly class. This involves tuning the classification threshold used by the model to decide whether a sample is anomalous. The process begins by establishing an acceptable miss rate that is aligned with business objectives. This

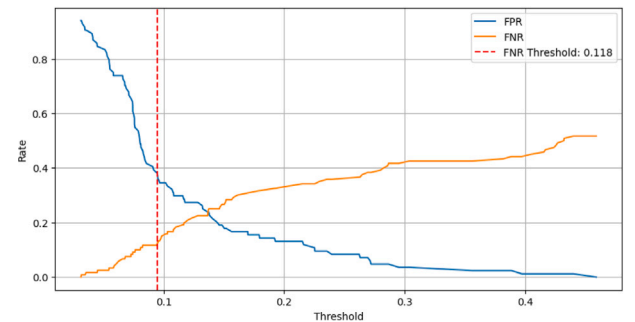


Fig. 3. Threshold optimization for EfficientAD: A threshold slightly above 10% FNR significantly reduces the FPR.

rate can be adjusted slightly higher or lower if significant performance improvements are observed. A baseline miss rate of 10% was defined for all models. To aid with the selection of this threshold, the False Negative Rate (FNR) can be plotted along with the False Positive Rate (FPR) across different thresholds. This allows for the selection of a threshold that balances these two metrics effectively. Fig. 3 illustrates a threshold optimization for EfficientAD, revealing that a slightly higher threshold than 10% significantly reduces the false positive rate, limiting the model's skewness towards the anomaly decision.

5.1.2. Evaluation metrics

In this section, we provide a brief description of the metrics chosen for evaluating the classification performance and anomaly localization of the selected models.

Image-wise metrics. The first metric is the Receiver Operating Characteristic Area Under the Curve (ROCAUC), which evaluates the model's

Table 2

Image-wise and Pixel-wise performance comparison on all tested models.

Model	Image-wise			Pixel-wise		
	AUC	F1-Score	P@TR	AUC	PRO	IoU
$\sigma = 2.0$	61.58	74.07	58.82	36.18	34.72	3.13
CFA	83.28	80.17	59.70	69.91	62.41	13.42
FastFlow	82.70	80.31	62.83	73.64	69.14	15.10
EfficientAD	85.54	82.35	60.30	75.72	75.14	18.10

ability to distinguish between classes. Additionally, the F1-score is used to balance precision and recall, particularly for imbalanced datasets. Precision at Total Recall (P@TR), or precision at 100% recall, indicates the proportion of correctly identified anomalies when all anomalies are retrieved.

Pixel-wise metrics. These metrics assess the correlation between the model's predictions and the ground truth defect segmentations. Similar to the earlier metrics, we calculate the AUC, but in this case at the pixel level. Additionally, we utilize Intersection over Union (IoU) and Per-Region Overlap AUC (PROAUC), which measure the AUC for the smallest connected component regions in the ground truth. The results of the benchmarks are shown in Table 2. These were obtained after the threshold optimization described in Section 5.1.1.

The Sigma-six (σ) approach is outperformed by the selected deep learning models on all of the benchmark metrics, proving that this approach is not well suited for the task at hand. In our experiments, $\sigma = 2$ proved to be the most robust standard deviation threshold. Values larger than this threshold resulted in a large number of images being classified as PASS. Even so, a $\sigma = 2$ threshold is too specific for the dataset's variability, making this solution unfeasible.

For image-wise metrics, EfficientAD achieves the best performance in terms of AUC and F1-scores. However, its P@TR is lower than that of FastFlow, indicating a tendency to overpredict anomalies at higher recall levels. Additionally, CFA is outperformed by all models in this evaluation. For pixel-wise metrics, EfficientAD emerges as the clear winner across all metrics, demonstrating higher confidence in pixel-level classification compared to other models. The PROAUC metric further highlights its superior ability to detect small defects. However, it is worth noting the low IoU scores observed across all models. This is primarily due to the ambiguity and the large number of DPC annotations, which are often inconsistently labeled as anomalies and are inherently subjective. Additionally, these models tend to prioritize severe outliers while overlooking smaller or redundant defects. As shown in Fig. 2, the anomaly detection model focused primarily on the most abnormal pads, assigning near-normal scores to the rest.

Fig. 2 illustrates the model's ability to localize the regions of an image affected by various defect types. Although some artifacts are scattered across the image, the highest intensity values are concentrated at the defect locations. The surrounding noisy areas further highlight the challenges the model faces in capturing the dataset's normality, stemming from the diverse structures present within it. A comparison of the anomaly maps of all the models can be found in Fig. D.11 to further highlight the choice of EfficientAD, over the other two. The confusion matrix in Fig. 4 provides deeper insight into the model's performance.

With threshold optimization, we observe a lower FNR at the cost of a slightly higher FPR, indicating some degree of overkill, as expected. The model exhibits heightened sensitivity to samples containing defects, even when they are not necessarily anomalies. This is particularly evident in images with small light particles, which are flagged as anomalies despite being easily manageable in practice. Further analysis revealed that 29 of the false positive images contained at least one type of defect, leading the model to classify them as anomalies. These results will be further analyzed in the results of the Dual-stage Output Fusion Approach (see Section 5.3).

Table 3

Class-wise performance comparison at pixel-level of the tested segmentation models.

Model	Pixel-level dice							mDice
	CON	DPC	DP	LP	MIS	OT	SAP	
SegFormer-B0	62.55	80.44	0	0	0	0	80.10	31.87
UPerNet-Swin	87.26	74.83	44.60	44.08	0	0	84.56	47.90
PIDNet	81.15	43.90	6.24	11.79	0	0	75.90	31.28

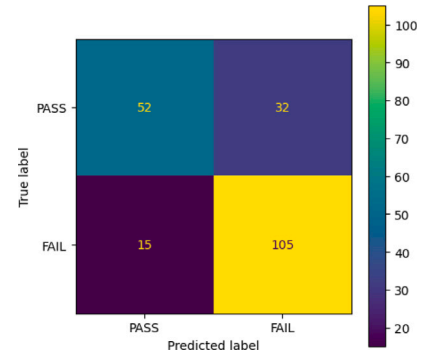


Fig. 4. Anomaly detection confusion matrix for EfficientAD. Class predictions were obtained with the classification threshold defined in Section 5.1.1.

5.2. Defect segmentation

To evaluate the segmentation performance of the investigated models, we employ both pixel-wise and mask-level metrics to provide a complete assessment of their effectiveness.

Pixel-wise metrics. The primary metric for segmentation performance is the pixel-wise Dice score, calculated for each of the seven distinct defect classes. The Dice score quantifies the overlap between the predicted segmentation mask and the ground truth, allowing us to capture class-specific segmentation quality. The results are presented in Table 3.

UPerNet-Swin achieves the highest mean Dice score (mDice) of 47.90, demonstrating superior robustness in handling a variety of defect types compared to SegFormer-B0 (31.87) and PIDNet (31.28). The CON, DPC, and SAP classes, characterized by larger and more distinct features, are more effectively segmented by all models, with UPerNet-Swin standing out. However, smaller and more localized defects such as DP and LP pose significant challenges. While PIDNet struggles with these classes and SegFormer-B0 fails to detect them entirely, UPerNet-Swin achieves Dice scores of approximately 44% for both, significantly outperforming the other models. This advantage can be attributed to its hierarchical transformer-based architecture, which captures multilevel contextual features, thereby enhancing segmentation accuracy for various types of defects. None of the models successfully segment the MIS and OT classes. This limitation is likely due to insufficient representation in the training dataset and the high variability inherent to the OT class. A comparison of the model outputs is provided in Fig. D.12 (see Fig. 5).

The confusion matrix analysis provides additional insights into the segmentation performance at both the multi-class and severity levels. In the multi-class evaluation, the model frequently confuses DP and LP, likely due to their similar small-scale features. Additionally, LP instances are occasionally misclassified as SAP, reflecting similarities in brightness and texture between these two classes. OT instances are often misidentified as either CON or LP, classes with distinct differences in appearance and size, which underscores the high intra-class variability of OT samples and the model's difficulty in learning to differentiate them. A strong bias towards background (BG) is also evident, driven

Table 4
Class-wise performance comparison at mask-level of the tested segmentation models.

Model	Mask-level metrics	Classes							wAvg.
		CON	DPC	DP	LP	MIS	OT	SAP	
SegFormer-B0	Recall	67.92	93.33	0	0	0	0	93.33	32.17
	Precision	45.00	56.00	0	0	0	0	45.90	19.03
	F1-Score	54.13	70.00	0	0	0	0	61.54	23.83
UPerNet-Swin	Recall	88.68	100	75.38	77.23	0	0	96.67	81.47
	Precision	62.67	25.21	44.95	39.20	0	0	69.05	45.56
	F1-Score	73.44	40.27	56.32	52.00	0	0	80.56	57.45
PIDNet	Recall	75.47	76.67	10.77	27.72	0	0	100	44.75
	Precision	26.14	17.29	11.48	14.14	0	0	17.05	16.05
	F1-Score	38.83	28.22	11.11	18.73	0	0	29.13	22.35

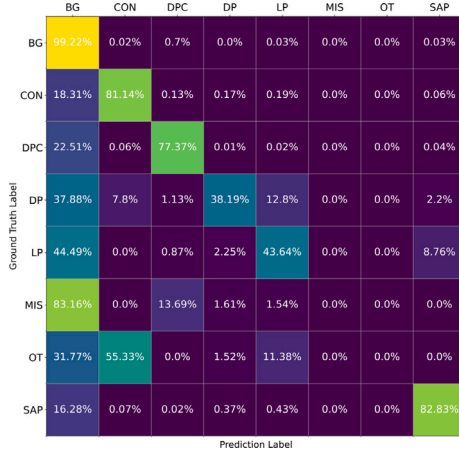


Fig. 5. Normalized confusion matrix for multi-class and aggregated classes approaches at pixel-level.

by its dominance in the dataset, which can overshadow the learning of minority classes.

Fig. 2 illustrates examples of predicted masks generated by the top performing model, UPerNet-Swin. Although the predictions may not always perfectly capture every pixel within the defect regions, the overall segmentation is satisfactory. However, as discussed earlier, the pixel-wise metrics may not fully reflect this performance. This suggests that a less stringent evaluation, focused on mask-level performance, may be more appropriate, as it emphasizes the broader accuracy of defect detection rather than demanding pixel-perfect segmentation.

Mask-level IoU-based metrics. To further enhance our evaluation, we perform segmentation performance comparison at a higher level, prioritizing overall mask agreement rather than exact pixel-by-pixel accuracy. In this approach, a prediction is considered correct if the Intersection over Union (IoU) score between the predicted and ground truth masks exceeds a predefined threshold. This approach prioritizes overall mask agreement, making it particularly useful in cases where minor pixel inaccuracies are tolerable. It emphasizes the model's ability to identify defects at the image level, regardless of perfect boundary alignment.

We compute true positives (TP), false positives (FP), and false negatives (FN) to derive recall and precision metrics, and subsequently the F1-score. Emphasizing recall is especially critical in defect detection contexts, as missing defects (false negatives) can lead to significant quality issues.

The results in Table 4 highlight the performance of the investigated models on class-wise mask-level metrics for defect detection. UPerNet-Swin achieves the highest weighted average recall (81.47%) and F1-score (57.45%), confirming its ability to identify multiple defect types. Its robust performance across both larger, more distinct defects

Table 5
Class-wise results of the dual-stage approach.

Class	Precision	Recall	F1-Score
PASS _{NONE}	0	0	0
PASS _{RECOV}	79.63	55.84	65.65
PASS _{DISCARD}	0	0	0
FAIL _{RECOV}	48.94	53.49	51.11
FAIL _{DISCARD}	80.00	93.51	86.23
wAvg.	70.57	67.65	68.10

like SAP and smaller, more localized defects like DP and LP underscores its capability to adapt to diverse defect patterns. This is consistent with its performance in pixel-level evaluations, where its hierarchical architecture enabled strong contextual feature learning. SegFormer-B0 achieves a low F1-score (23.83%) and recall (32.17%), effectively handling certain classes like DPC but struggling with rare or small defects such as DP and LP. PIDNet, with the lowest average F1-score (22.35%), shows limitations in balancing recall and precision, highlighting its tendency to over-predict defective regions while underperforming in accurately segmenting small or rare defects. As observed in the pixel-level evaluation, none of the models effectively segment MIS and OT classes at the mask level. These classes remain problematic due to their high intra-class variability and insufficient representation in the training dataset. The confusion between DP and LP classes, likely due to their similar small-scale features, is also reflected in the mask-level performance, with consistently low F1-scores across models for these classes.

Overall, UPerNet-Swin emerges as the most effective model, excelling in balancing recall and F1-score for both large, distinct defects and smaller, more localized ones. However, it presents limitations in segmenting rare classes like MIS and OT and the persistent challenges in distinguishing similar small-scale defects such as DP and LP.

5.3. Dual-stage output fusion approach

This section presents the results of the non-trainable dual-stage approach, which combines the best-performing methods for unsupervised anomaly detection (EfficientAD) and supervised defect detection (UPerNet-Swin). The results of this dual-stage approach are summarized in Table 5 and the confusion matrix is presented in Fig. 6.

Analyzing the results, it can be inferred from the high decision reliability from three categories - PASS_{NONE}, PASS_{RECOV}, FAIL_{DISCARD} - that these require no operator intervention. The system's performance in detecting FAIL_{DISCARD} samples is noteworthy, with a recall of 93.51%, i.e., 72 out of 77 true instances being correctly identified. This demonstrates a strong capability to detect wafers with severe defects requiring immediate disposal, ensuring the system's critical function of minimizing the risk of defective wafers proceeding further in the process. Moreover, the system did not incorrectly label defective wafers as defect-free. This is a critical outcome, as it mitigates the risk of false

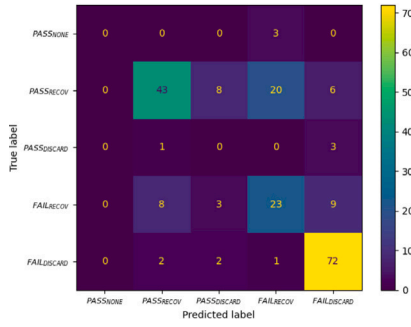


Fig. 6. Confusion matrix for the decision obtained by the dual-stage approach.

negatives, where defects could be overlooked. For $PASS_{RECOV}$, severe defects are found in only 14 cases, while 63 wafers are deemed as recoverable. Despite the $PASS_{NONE}$ class having F1-score of 0%, due to its three instances being classified into more severe categories by our dual-stage approach, we maintain that no human intervention is necessary. This outcome aligns with the system's objective of prioritizing defect detection, ensuring that no defective wafers are mistakenly classified as defect-free. In contrast, cases labeled as $FAIL_{RECOV}$ require manual operator review, showcased by a recall value of 53.49% and precision of 48.94%, with only 23 out of 43 instances were correctly classified. Misclassifications in this category were evenly split between $PASS_{RECOV}$ and $FAIL_{DISCARD}$, reflecting difficulties in distinguishing between recoverable and non-recoverable defects. This issue likely stems from overlapping feature characteristics in the segmentation model, where subtle differences in defect appearance challenge the model's ability to consistently differentiate defect severity. Moreover, the complexity of the $PASS_{DISCARD}$ category is highlighted. A total of 13 samples were incorrectly assigned to this class, while its 4 ground truth instances were misclassified (1 as $PASS_{RECOV}$ and 3 as $FAIL_{DISCARD}$). The challenge arises because these samples include defects classified as severe (CON and SAP), yet the expert annotators still consider these wafers as PASS. This occurs when defects are small in size or have lower pixel intensity, allowing them to be considered as recoverable cases. However, since our dual-stage approach does not explicitly account for these two parameters, some borderline instances are misclassified.

The dual-stage approach effectively mitigates the anomaly detector's sensitivity to small deviations from the normality. By integrating defect segmentation outputs, the system can refine its decisions, allowing for more nuanced classification between recoverable and discardable wafers. On the other hand, supervised segmentation relies on labeled defect types, which in the wafer production process is inherently dynamic and prone to rare or previously unseen defects that are difficult to capture exhaustively through annotation. By integrating an unsupervised anomaly detection stage, we introduce a complementary mechanism capable of flagging unannotated or novel defects, which might otherwise be missed by the supervised model. This hybrid approach enhances robustness in high-throughput manufacturing settings by enabling early detection of unforeseen anomalies, while still allowing the operator to classify or investigate new defect patterns as they arise.

5.4. Computational efficiency analysis

All experiments were conducted on NVIDIA® V100 GPU (virtualized with 16 GB of VRAM) and an Intel® Xeon® Silver 4114 CPU. As shown in Table 6, CFA achieves the lowest inference time in the anomaly detection step. However, EfficientAD is considered the most suitable model for this task, as it delivers significantly higher performance in the anomaly detection benchmark.

In Defect Segmentation, the UPerNet-Swin model has twice the inference time of the other segmentation models, but its significantly

Table 6

Computational efficiency of the different explored approaches.

Approach	Model	Inference time (ms)
Anomaly detection	CFA	7.25 ± 1.32
	FastFlow	28.65 ± 2.85
	EfficientAD	10.56 ± 0.85
Defect segmentation	SegFormer-B0	21.39 ± 1.42
	UPerNet-Swin	45.10 ± 0.79
	PIDNet	22.25 ± 1.36
Dual-Stage	EfficientAD + UPerNet-Swin	55.66 ± 1.64

superior performance and acceptable processing time make it the preferred choice for both Defect Segmentation and Dual-Stage modules, as the benefits outweigh the computational cost. These comparative results demonstrate that all three approaches exhibit competitive computational efficiency, making them well-suited for deployment in real-world settings with high-throughput demands.

6. Conclusions and future work

Current advancements in AI offer a promising solution to the increasing complexity of defect detection in semiconductor manufacturing. Our study highlights the effectiveness of novel lightweight AI models in enhancing automated wafer visual inspection, leveraging a newly annotated dataset of 1055 wafer images. From a data-centric perspective, our novel unsupervised dataset-splitting approach ensures balanced representation across defect types, achieving 96% coverage while maintaining the target test ratio. From a model-centric perspective, benchmarking recent methods reveals that EfficientAD excels in unsupervised anomaly detection, while UPerNet-Swin delivers the best performance in supervised defect segmentation. Additionally, the explored non-trainable dual-stage output fusion approach enhances PASS/FAIL decisions by incorporating defect severity, bridging the gap between anomaly detection and defect segmentation. Finally, our comparative analysis of computational efficiency highlights the feasibility of deploying these models in high-throughput environments.

While the current system demonstrates strong potential for automated wafer defect classification, further refinement of the segmentation module is necessary to address challenges in ambiguous cases such as $FAIL_{RECOV}$. Incorporating defect size and intensity thresholds in future iterations could improve the system's ability to distinguish between recoverable and non-recoverable defects with greater accuracy.

Despite the absence of pristine images in the dataset and the high prevalence of defects in every sample, the anomaly detection model has proven effective in generating reliable classifications. However, many false positive anomalies correspond to minor defects that, while not severe, should still be flagged for further evaluation by the segmentation model. Improving segmentation performance for smaller and more localized defects, such as DP and LP, remains a priority. Additionally, the inability of all models to effectively detect MIS and OT defects across evaluation levels highlights dataset-level challenges, including insufficient representation and high intra-class variability. Further research could include advanced data augmentation strategies, such as generative AI, and loss weighting strategies to mitigate the impact of limited data representation and enhance segmentation performance.

Finally, future work on the dual-stage approach could explore additional trainable fusion strategies and benchmark them against the current non-trainable output fusion method. Potential approaches include feature-level fusion techniques that integrate extracted feature maps from both models using concatenation or attention mechanisms, followed by processing through a final classifier head. These strategies could enhance the synergy between anomaly detection and defect segmentation, potentially improving overall automated wafer inspection performance.

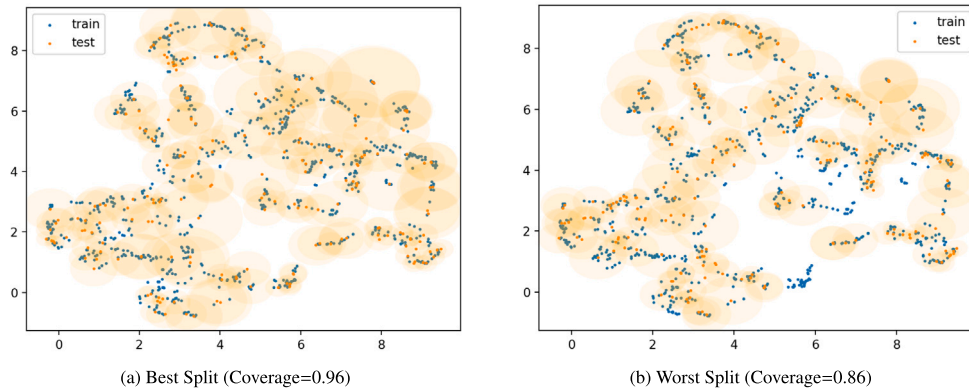


Fig. A.7. Coverage visualization for best (left) and worst (right) splits. This is a UMAP 2D reduction of the features extracted with DINO-ViT-S/8 on the AOI dataset. Each point represents an image in the dataset, where the train and test instances are represented in blue and orange, respectively. Additionally, the coverage of each sample, as defined in Improved-Recall (Kynkäänniemi et al., 2019), is also shown through circumferences with the center in the sample and the radius corresponding to its k th nearest-neighbor.

Table B.9

Image-wise and Pixel-wise performance comparison under random and representative splits.

Split	Image-wise			Pixel-wise		
	AUC	F1-Score	P@TR	AUC	PRO	IoU
Random	77.36 $\pm$ 3.76	77.33 $\pm$ 1.27	60.10 $\pm$ 1.29	67.39 $\pm$ 2.22	66.10 $\pm$ 3.26	13, 56 $\pm$ 3.01
Repr. Split	79.35 $\pm$ 2.50	78.45 $\pm$ 1.84	59.58 $\pm$ 0.59	68, 12 $\pm$ 1.20	66.91 $\pm$ 1.71	13.91 $\pm$ 2.08

CRedit authorship contribution statement

Ivo Façoco: Conceptualization, Methodology, Implementation, Anomaly Detection, Results analysis, Validation, Writing – original draft, Writing – review & editing. **Rafaela Carvalho:** Conceptualization, Methodology, Implementation, Defect segmentation, Results analysis, Validation, Writing – original draft, Writing – review & editing. **Luís Rosado:** Conceptualization, Methodology, Results analysis, Validation, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

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Appendix A. Representative split report

You can find the split reports in [Tables A.7](#) and [A.8](#).

Appendix B. Dataset splitter tests

We conducted a controlled experiment using the best-performing model from the Anomaly Detection stage (EfficientAD) to evaluate the effectiveness of the data splitter. The goal is to assess whether the representative splitter produces validation sets that better reflect the characteristics of the test set. We created three data splits, each using 10% of the training set, comparing both random and representative sampling strategies across different seeds. For each split, the EfficientAD model was retrained, and the validation set was used to select the best-performing checkpoint. Finally, the selected models were evaluated on the original test set to compare generalization performance. The results are expressed in [Table B.9](#).

Table A.7

Test ratio per defect type outputted by the representative splitter. Some classes have slightly lower proportions than the defined 0.2 for the test set, due to the constraints applied in the splitting of the dataset.

Defect	Test ratio
LP	0.185
CON	0.195
DP	0.221
DPC	0.205
SAP	0.213
OT	0.206
MIS	0.185
NONE	0.2

Table A.8

Test ratio per defect type per decision outputted by the representative splitter.

Defect/Decision	Test ratio	
	PASS	FAIL
LP	0.182	0.192
CON	0.152	0.200
DP	0.207	0.254
DPC	0.209	0.204
SAP	0.167	0.214
OT	0.200	0.207
MIS	0.154	0.194
NONE	–	0.200

The results in [Table B.9](#) show that EfficientAD models trained using representative splits consistently achieve slightly better performance than those trained on randomly selected splits. Additionally, the random strategy exhibits greater variability across runs, suggesting that the representative splitter contributes to more stable and reliable model training.

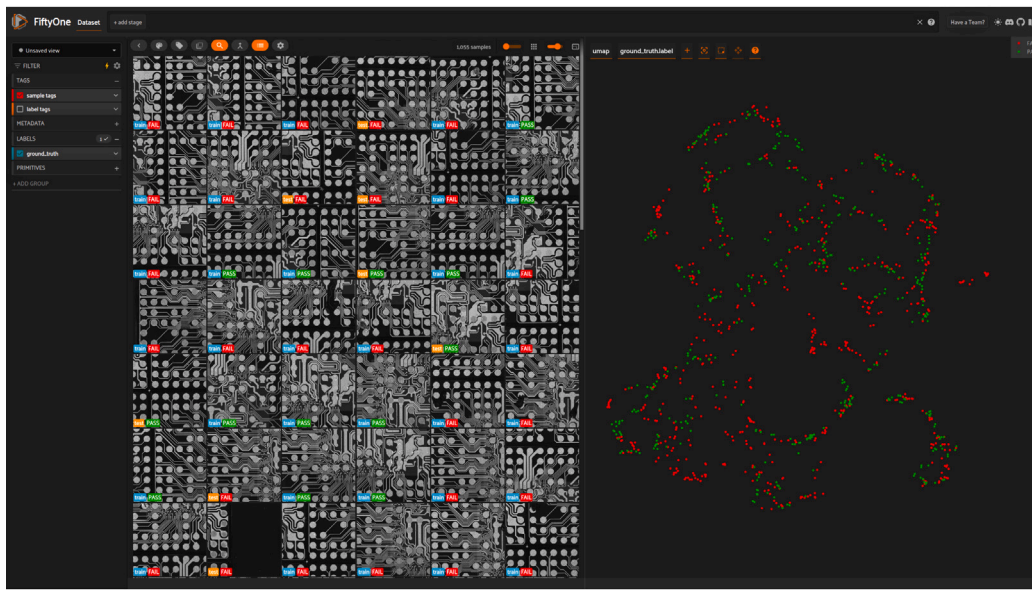


Fig. C.8. FiftyOne visualization of dataset and image features per decision class. FAIL and PASS images are identified in red and green, respectively.

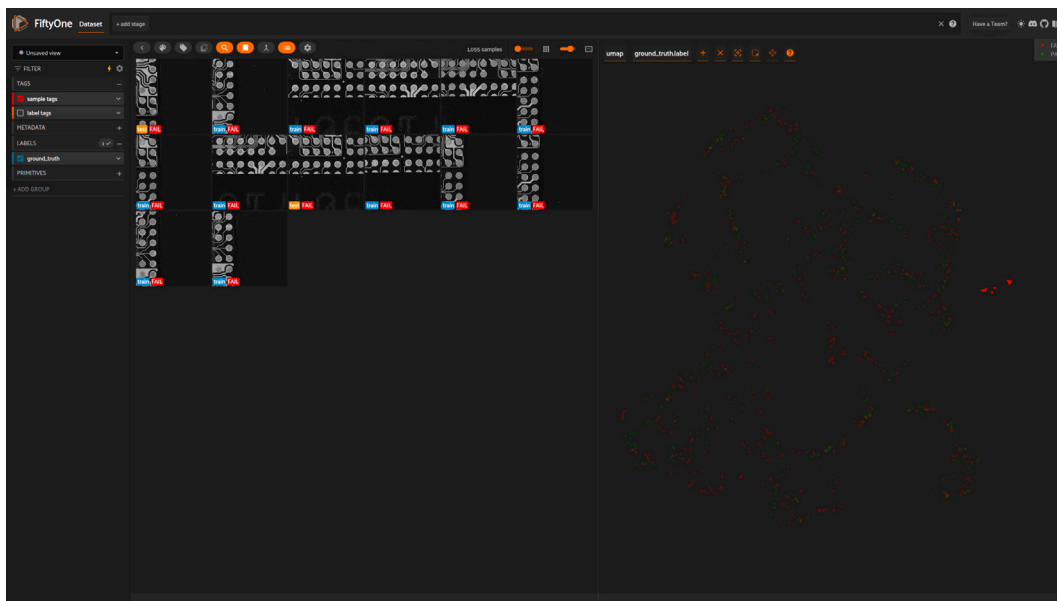


Fig. C.9. FiftyOne selection of FAIL samples. Selection of FAIL samples through the embedding visualization interface. The images in this cluster present similar features.

Appendix C. Dataset visualization tool

In addition to the splitter tool, we developed a visualization tool using FiftyOne (Moore & Corso, 2020) to explore the dataset's characteristics in greater detail and ensure that the variety of images in the dataset is adequately represented in the test set. The interface comprises two main components designed to aid in debugging the splitting behavior and identifying common characteristics within the dataset. The left panel provides a condensed view of the dataset images and their labels, which can represent either train/test splits or PASS/FAIL image-wise decisions. The right component features an interactive version of the

plot shown in Fig. A.7, allowing users to select specific clusters and view the corresponding images in the left panel (see Figs. C.8–C.10).

Appendix D. Model output comparisons

Figs. D.11 and D.12 showcase the tradeoffs between the considered anomaly and defect detection solutions.

Data availability

The authors do not have permission to share data.

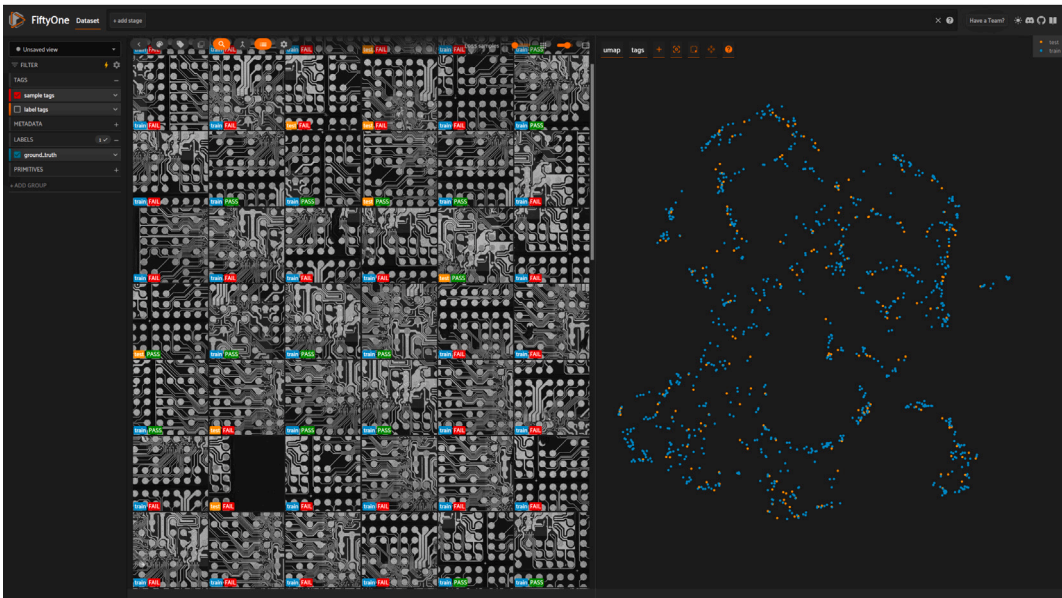


Fig. C.10. FiftyOne visualization of dataset and image features per split label. Train and test images are represented in blue and green, respectively.

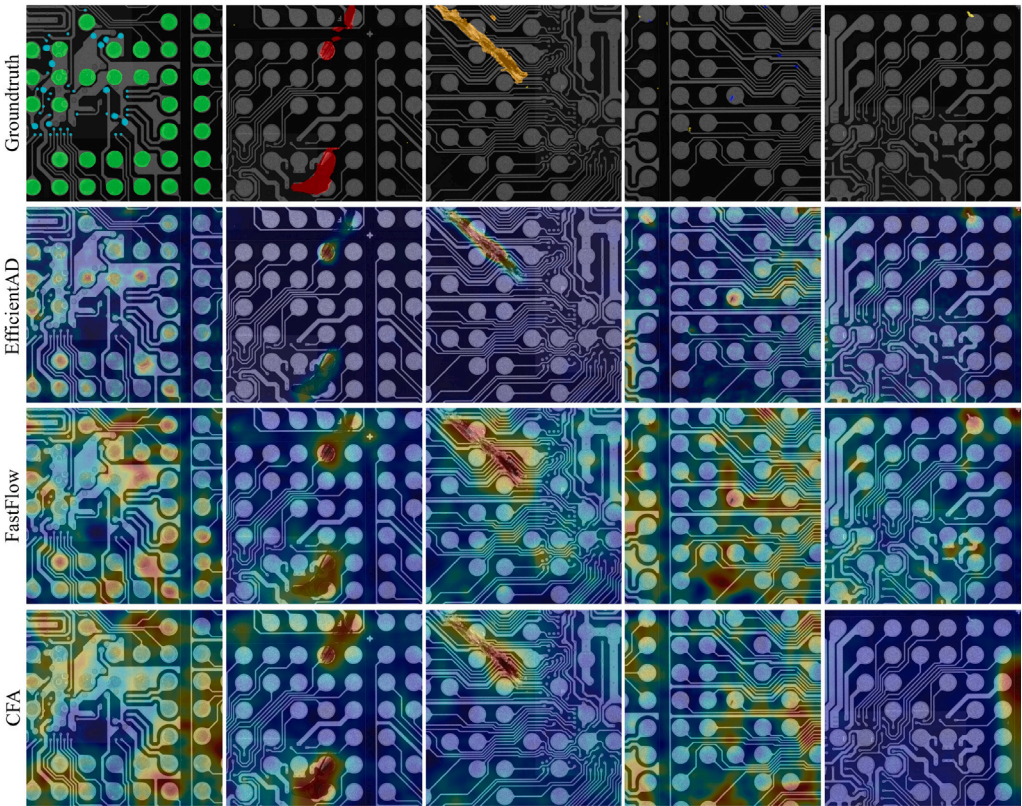


Fig. D.11. Comparison of the outputs produced by the tested deep learning models. Of the three models, EfficientAD is the one that produces clearer maps, while the other are noisier.

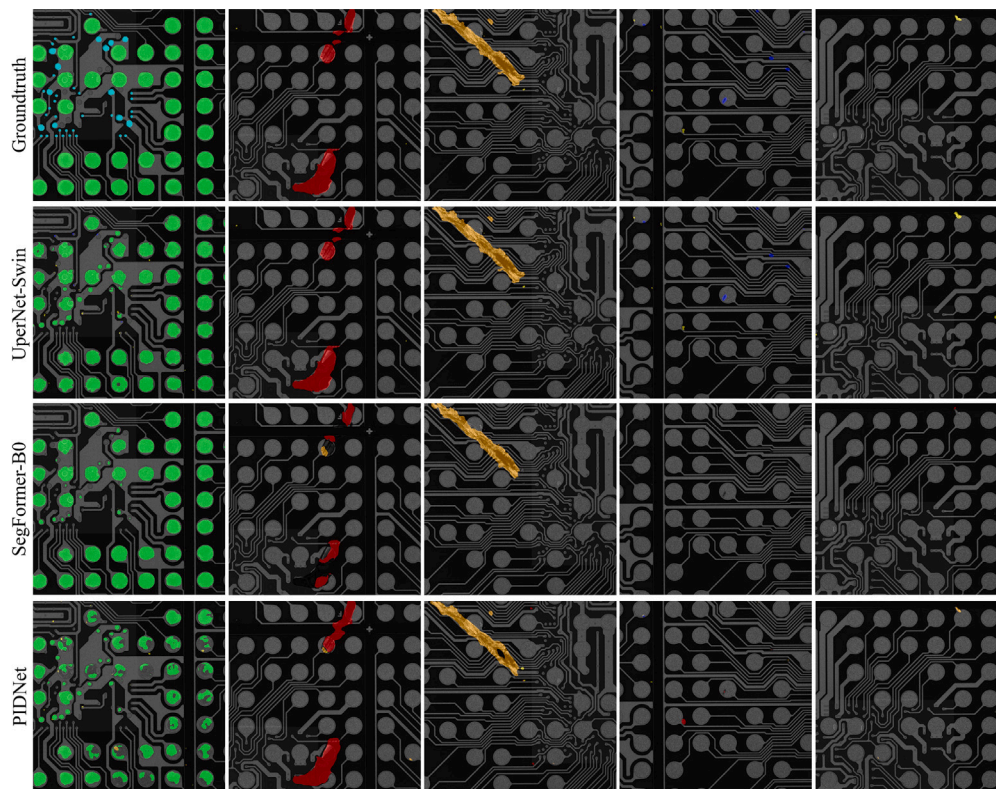


Fig. D.12. Comparison of the outputs produced by the tested deep learning segmentation models. UperSwinNet produces the most robust pixel level classifications, even being able to detect the smallest of defects, while in SegFormer or PIDNet, these are misclassified or not classified at all.

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